

# **Digital Electronics**

**ECE 2105**

# SYLLABUS

**Introduction of Combinational logic design:** Overview of Boolean Algebra and K-Map, Half and Full Adders, Subtractors, Serial and Parallel Adders, BCD Adder. MSI devices: Comparators, Multiplexers, Encoder, Decoder, Driver & Multiplexed Display, Barrel shifter and ALU.

**Sequential logic design:** latch, Flip-flop, S-R FF, D FF, JK FF, T FF, and Master-Slave JK FF, Edge triggered FF, Ripple and Synchronous counters, Shift registers, Timing Analysis of sequential circuits.

**Designing of State Machines:** Finite state machines, Design of synchronous FSM, State Reduction, Timing issues in synchronous circuits. Algorithmic State Machines, Designing synchronous circuits like Pulse train generator, Pseudorandom Binary Sequence generator, Clock generation. Design of asynchronous circuits.

**Logic Families and Semiconductor Memories:** TTL NAND gate, Specifications, Noise margin, Propagation delay, fan-in, fan-out, Tristate TTL, ECL, CMOS families and their interfacing, Memory elements, Concept of Programmable logic devices, Logic implementation using Programmable Devices.

## **TEXT BOOKS:**

1. A. A. Kumar, FUNDAMENTALS OF DIGITAL CIRCUITS. Prentice Hall India Pvt., Limited, (2e), 2016.
2. R. P. Jain, Modern Digital Electronics. McGraw-Hill Education (India) Pvt Limited, (4e), 2003.
3. W.H. Gothmann, Digital Electronics- An introduction to theory and practice, PHI, (2e), 2006.
4. S. Brown and Z. Vranesic, Fundamentals of Digital logic with Verilog Design, McGraw Hill, (3e) 2013.
5. M. Mano and M. Ciletti, “Digital Design: With an introduction to Verilog HDL”, Pearson, 2013.

# Overview

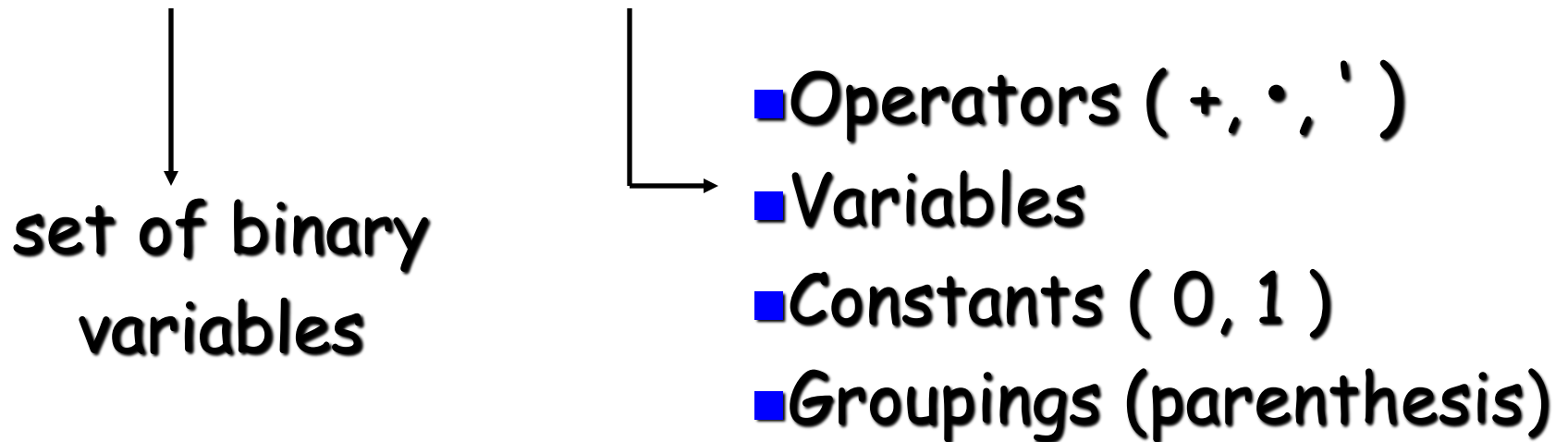
- Boolean Algebra
  - Basic Properties
  - Algebraic Manipulation
- Standard and Canonical Forms
  - Minterms and Maxterms (Canonical forms)
  - SOP and POS (Standard forms)
- Binary logic and Gates

# Binary Logic

- Deals with binary variables that take 2 discrete values (0 and 1), and with logic operations
- Three basic logic operations:
  - AND, OR, NOT
- Binary/logic variables are typically represented as letters: A,B,C,...,X,Y,Z

# Binary Logic Function

$F(\text{vars}) = \text{expression}$

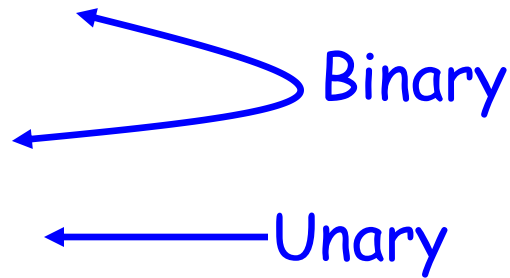


Example:  $F(a,b) = a' \cdot b + b'$

$G(x,y,z) = x \cdot (y + z')$

# Basic Logic Operators

- AND
- OR
- NOT



- $F(a,b) = a \bullet b$ ,  $F$  is 1 if and only if  $a=b=1$
- $G(a,b) = a + b$ ,  $G$  is 1 if either  $a=1$  or  $b=1$
- $H(a) = a'$ ,  $H$  is 1 if  $a=0$

# Basic Logic Operators (cont.)

- 1-bit logic AND resembles binary multiplication:

$$0 \bullet 0 = 0, \quad 0 \bullet 1 = 0,$$

$$1 \bullet 0 = 0, \quad 1 \bullet 1 = 1$$

- 1-bit logic OR resembles binary addition, except for one operation:

$$0 + 0 = 0, \quad 0 + 1 = 1,$$

$$1 + 0 = 1, \quad 1 + 1 = 1 \ (\neq 10_2)$$

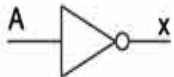
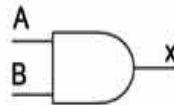
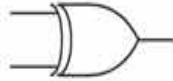


**Que.:** Let a function  $F()$  depend on  $n$  variables. How many cases are there in the truth table of  $F()$  ?

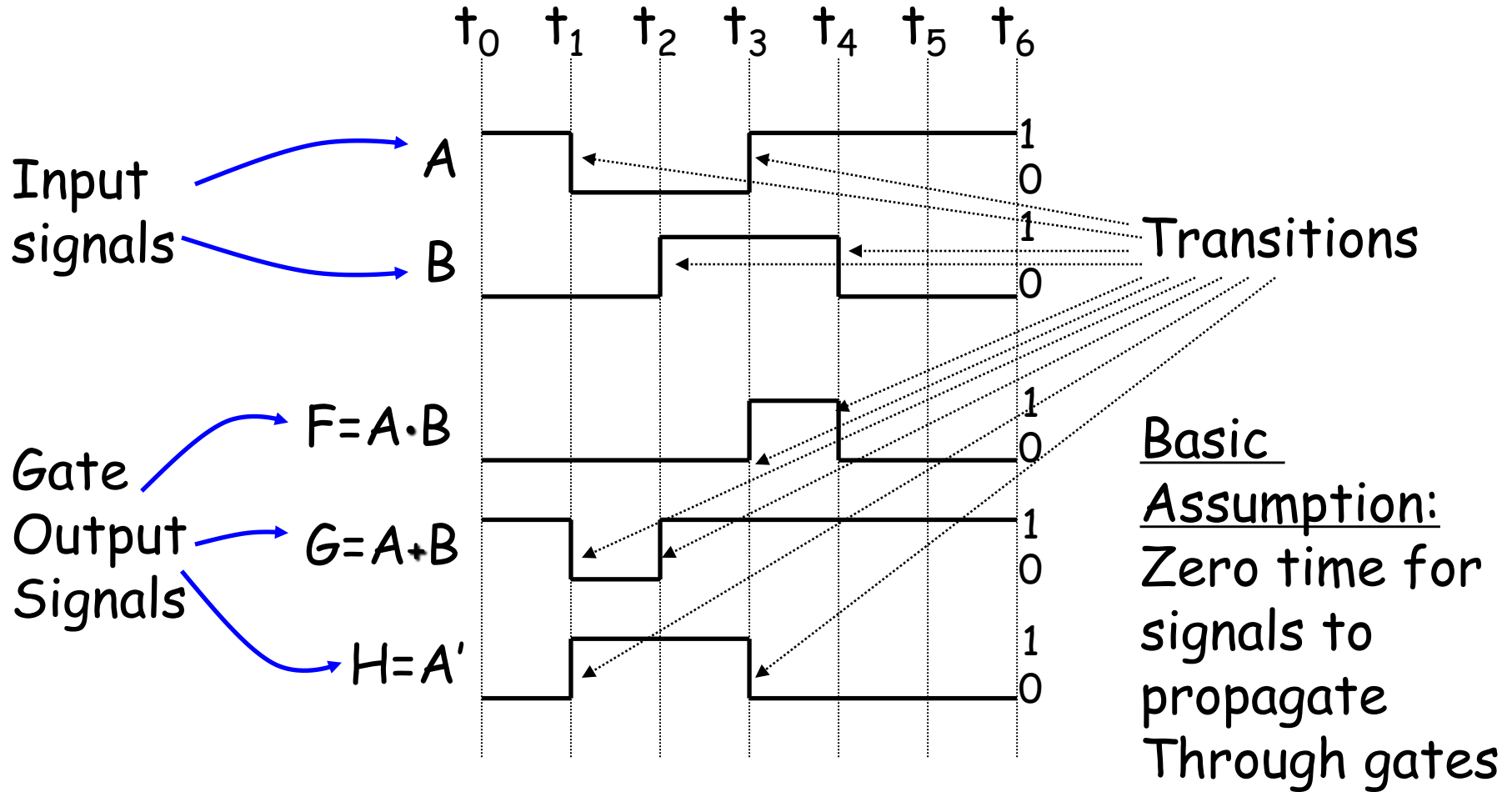
**Ans.:**  $2^n$  rows, since there are  $2^n$  possible binary patterns/combinations for the  $n$  variables

# Logic Gates

- Logic gates are abstractions of electronic circuit components that operate on one or more input signals to produce an output signal.

| Name        | NOT                                                                                                    | AND                                                                               | NAND                                                                              | OR                                                                                 | NOR                                                                                 | XOR                                                                                 | XNOR                                                                                |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|-------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Alg. Expr.  | $\overline{A}$                                                                                         | $AB$                                                                              | $\overline{AB}$                                                                   | $A+B$                                                                              | $\overline{A+B}$                                                                    | $A\oplus B$                                                                         | $\overline{A\oplus B}$                                                              |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Symbol      |                       |  |  |  |  |  |  |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Truth Table | <table><tr><th>A</th><th>X</th></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr></table> | A                                                                                 | X                                                                                 | 0                                                                                  | 1                                                                                   | 1                                                                                   | 0                                                                                   | <table><tr><th>B</th><th>A</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table> | B | A | X | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | <table><tr><th>B</th><th>A</th><th>X</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table> | B | A | X | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | <table><tr><th>B</th><th>A</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table> | B | A | X | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 1 | <table><tr><th>B</th><th>A</th><th>X</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table> | B | A | X | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | <table><tr><th>B</th><th>A</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table> | B | A | X | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | <table><tr><th>B</th><th>A</th><th>X</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table> | B | A | X | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 1 |
| A           | X                                                                                                      |                                                                                   |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0           | 1                                                                                                      |                                                                                   |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1           | 0                                                                                                      |                                                                                   |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| B           | A                                                                                                      | X                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0           | 0                                                                                                      | 0                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0           | 1                                                                                                      | 0                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1           | 0                                                                                                      | 0                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1           | 1                                                                                                      | 1                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| B           | A                                                                                                      | X                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0           | 0                                                                                                      | 1                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0           | 1                                                                                                      | 1                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1           | 0                                                                                                      | 1                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1           | 1                                                                                                      | 0                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| B           | A                                                                                                      | X                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0           | 0                                                                                                      | 0                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0           | 1                                                                                                      | 1                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1           | 0                                                                                                      | 1                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1           | 1                                                                                                      | 1                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| B           | A                                                                                                      | X                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0           | 0                                                                                                      | 1                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0           | 1                                                                                                      | 0                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1           | 0                                                                                                      | 0                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1           | 1                                                                                                      | 0                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| B           | A                                                                                                      | X                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0           | 0                                                                                                      | 0                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0           | 1                                                                                                      | 1                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1           | 0                                                                                                      | 1                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1           | 1                                                                                                      | 0                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| B           | A                                                                                                      | X                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0           | 0                                                                                                      | 1                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0           | 1                                                                                                      | 0                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1           | 0                                                                                                      | 0                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1           | 1                                                                                                      | 1                                                                                 |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

# Timing Diagram

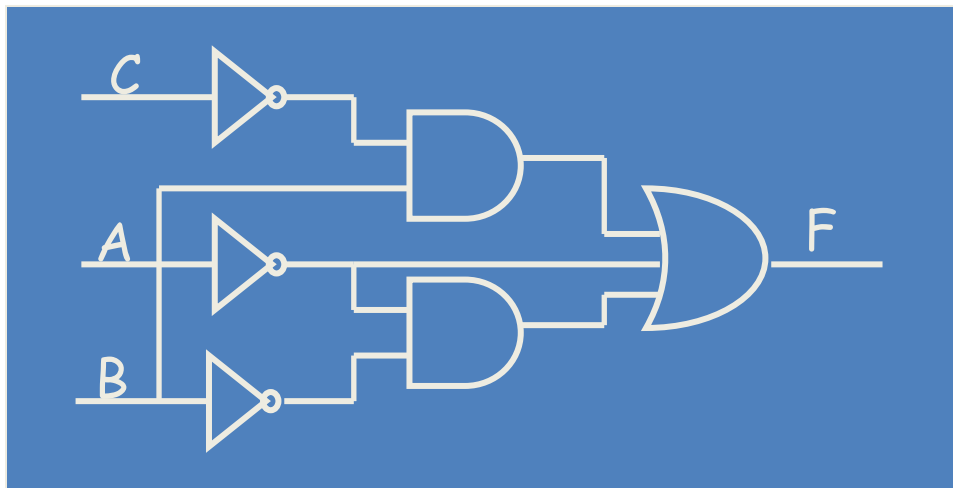


# Logic Circuit from Logic Function

- Consider function  $F = A' + B \cdot C' + A' \cdot B'$

A combinational logic circuit can be constructed to implement  $F$ , by appropriately connecting input signals and logic gates:

- Circuit input signals  $\rightarrow$  from function variables ( $A, B, C$ )
- Circuit output signal  $\rightarrow$  function output ( $F$ )
- Logic gates  $\rightarrow$  from logic operations



# Combinational Logic Circuit from Logic Function (cont.)

- In order to design a cost-effective and efficient circuit, we must minimize the circuit's size (area) and propagation delay (time required for an input signal change to be observed at the output line)

- Observe the truth table of

$$F = A' + B \cdot C' + A' \cdot B'$$

$$F = A' (1 + B') + B \cdot C'$$

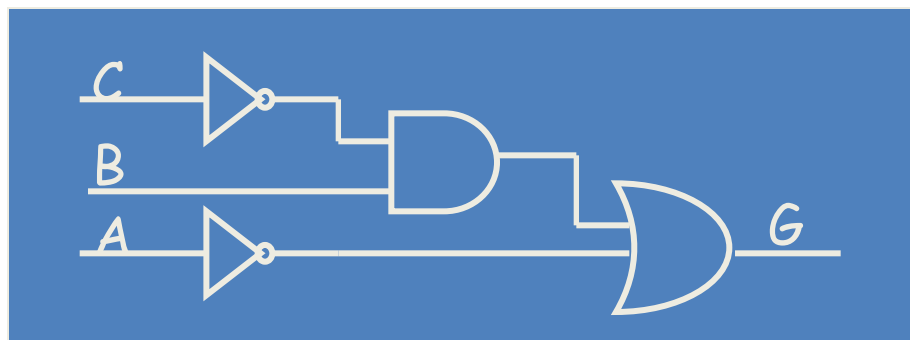
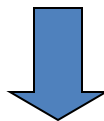
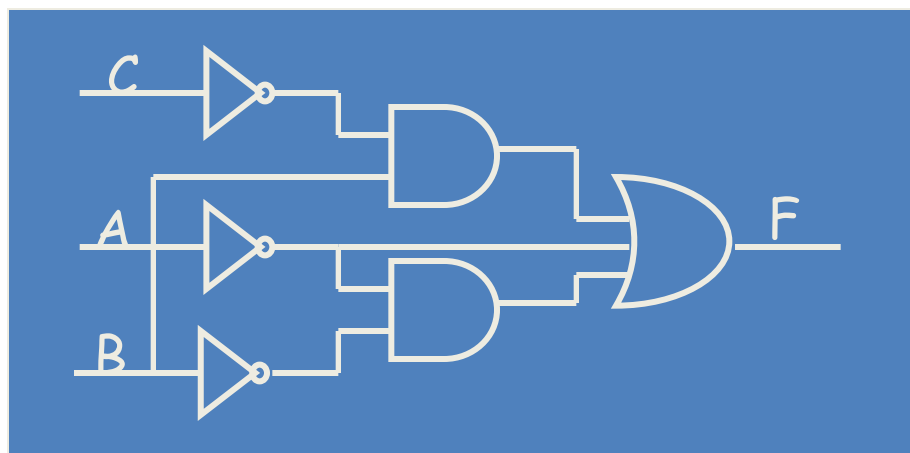
$$G = A' + BC'$$

Truth tables for F and G are identical →  
same function

- Use G to implement the logic circuit (less components)

| A | B | C | F | G |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 1 | 1 | 1 |
| 0 | 1 | 0 | 1 | 1 |
| 0 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 0 |
| 1 | 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 0 | 0 |

## Combinational Logic Circuit from Logic Function (cont.)



# Boolean Algebra Properties

Let X: Boolean variable, 0,1: constants

1.  $X + 0 = X$  -- Zero Axiom
2.  $X \cdot 1 = X$  -- Unit Axiom
3.  $X + 1 = 1$  -- Unit Property
4.  $X \cdot 0 = 0$  -- Zero Property
5.  $X + X = X$  -- Idepotence
6.  $X \cdot X = X$  -- Idepotence
7.  $X + X' = 1$  -- Complement
8.  $X \cdot X' = 0$  -- Complement
9.  $(X')' = X$  -- Involution
10.  $X + Y = Y + X$
11.  $X \cdot Y = Y \cdot X$  -- Commutative
12.  $X + (Y + Z) = (X + Y) + Z$
13.  $X \cdot (Y \cdot Z) = (X \cdot Y) \cdot Z$  -- Associative
14.  $X \cdot (Y + Z) = X \cdot Y + X \cdot Z$
15.  $X + (Y \cdot Z) = (X + Y) \cdot (X + Z)$  -- Distributive
16.  $(X + Y)' = X' \cdot Y'$
17.  $(X \cdot Y)' = X' + Y'$  -- DeMorgan's

And many more...

# Truth Tables (cont.)

- Truth table: a unique representation of a Boolean function
- Tabular form that represents the relationship between the input variables of a function and its output
- If two functions have identical truth tables, the functions are equivalent (and vice-versa).
- Truth tables can be used to prove equality theorems.
- However, the size of a truth table grows exponentially with the number of variables involved, hence unwieldy. This motivates the use of Boolean Algebra.



# Boolean expressions-NOT unique

- Unlike truth tables, expressions representing a Boolean function are NOT unique.
- Example:
  - $F(x,y,z) = x' \cdot y' \cdot z' + x' \cdot y \cdot z' + x \cdot y \cdot z'$
  - $G(x,y,z) = x' \cdot y' \cdot z' + y \cdot z'$
- The corresponding truth tables for  $F(x,y,z)$  and  $G(x,y,z)$  are to the right. They are identical.
- Thus,  $F(x,y,z) = G(x,y,z)$

| x | y | z |  | F | G |
|---|---|---|--|---|---|
| 0 | 0 | 0 |  | 1 | 1 |
| 0 | 0 | 1 |  | 0 | 0 |
| 0 | 1 | 0 |  | 1 | 1 |
| 0 | 1 | 1 |  | 0 | 0 |
| 1 | 0 | 0 |  | 0 | 0 |
| 1 | 0 | 1 |  | 0 | 0 |
| 1 | 1 | 0 |  | 1 | 1 |
| 1 | 1 | 1 |  | 0 | 0 |

# Algebraic Manipulation

- Boolean algebra is a useful tool for simplifying digital circuits.
- Why do it? Simpler can mean cheaper, smaller, faster.
- Example: Simplify  $F = x'yz + x'yz' + xz$ .

$$\begin{aligned} F &= x'yz + x'yz' + xz \\ &= x'y(z+z') + xz \\ &= x'y \cdot 1 + xz \\ &= x'y + xz \end{aligned}$$

# Algebraic Manipulation (cont.)

- Example: Prove

$$x'y'z' + x'yz' + xyz' = x'z' + yz'$$

- **Proof:**

$$\begin{aligned} x'y'z' + x'yz' + xyz' \\ &= x'y'z' + x'yz' + x'yz' + xyz' \\ &= x'z'(y' + y) + yz'(x' + x) \\ &= x'z' \cdot 1 + yz' \cdot 1 \\ &= x'z' + yz' \end{aligned}$$

# Canonical and Standard Forms

- We need to consider formal techniques for the simplification of Boolean functions.
  - Identical functions will have exactly the same canonical form.
  - Minterms and Maxterms
  - Sum-of-Minterms and Product-of- Maxterms
  - Product and Sum terms
  - Sum-of-Products (SOP) and Product-of-Sums (POS)

# Definitions

- *Literal*: A variable or its complement
- *Product term*: literals connected by  $\cdot$
- *Sum term*: literals connected by  $+$
- *Minterm*: a product term in which all the variables appear exactly once, either complemented or actual literal
- *Maxterm*: a sum term in which all the variables appear exactly once, either complemented or actual literal.

# Minterm

- Represents exactly one combination in the truth table.
- Denoted by  $m_j$ , where  $j$  is the decimal equivalent of the minterm's corresponding binary combination ( $b_j$ ).
- A variable in  $m_j$  is complemented if its value in  $b_j$  is 0, otherwise its value in  $b_j$  is 1.
- Example: Assume 3 variables (A,B,C), and  $j=3$ . Then,  $b_j = 011$  and its corresponding minterm is denoted by

$$m_j = A'BC$$

# Maxterm

- Represents exactly one combination in the truth table.
- Denoted by  $M_j$ , where  $j$  is the decimal equivalent of the maxterm's corresponding binary combination ( $b_j$ ).
- A variable in  $M_j$  is complemented if its value in  $b_j$  is 1, otherwise its value in  $b_j$  is 0.
- Example: Assume 3 variables (A,B,C), and  $j=3$ . Then,  $b_j = 011$  and its corresponding maxterm is denoted by

$$M_j = A+B'+C'$$

# Truth Table notation for Minterms and Maxterms

- Minterms and Maxterms are easy to denote using a truth table.
- Example:  
Assume 3 variables x,y,z  
(order is fixed)

| x | y | z | Minterm        | Maxterm          |
|---|---|---|----------------|------------------|
| 0 | 0 | 0 | $x'y'z' = m_0$ | $x+y+z = M_0$    |
| 0 | 0 | 1 | $x'y'z = m_1$  | $x+y+z' = M_1$   |
| 0 | 1 | 0 | $x'yz' = m_2$  | $x+y'+z = M_2$   |
| 0 | 1 | 1 | $x'yz = m_3$   | $x+y'+z' = M_3$  |
| 1 | 0 | 0 | $xy'z' = m_4$  | $x'+y+z = M_4$   |
| 1 | 0 | 1 | $xy'z = m_5$   | $x'+y+z' = M_5$  |
| 1 | 1 | 0 | $xyz' = m_6$   | $x'+y'+z = M_6$  |
| 1 | 1 | 1 | $xyz = m_7$    | $x'+y'+z' = M_7$ |



# Canonical Forms (Unique)

- Any Boolean function  $F()$  can be expressed as a *unique* **sum** of **minterms** and a unique **product** of **maxterms** (under a fixed variable ordering).
- In other words, every function  $F()$  has two canonical forms:
  - Canonical Sum-Of-Products (sum of minterms)
  - Canonical Product-Of-Sums (product of maxterms)

# Canonical Forms (cont.)

- Canonical Sum-Of-Products:

The minterms included are those  $m_j$  such that  $F() = 1$  in row  $j$  of the truth table for  $F()$ .

$$F = ABC + ABC' + AB'C'$$

- Canonical Product-Of-Sums:

The maxterms included are those  $M_j$  such that  $F() = 0$  in row  $j$  of the truth table for  $F()$ .

$$F = (A' + B' + C')(A' + B' + C)(A' + B + C)$$

# Example

- Truth table for  $f_1(a,b,c)$  at right
- The canonical sum-of-products form for  $f_1$  is

$$\begin{aligned} f_1(a,b,c) &= m_1 + m_2 + m_4 + m_6 \\ &= a'b'c + a'bc' + ab'c' + abc' \end{aligned}$$

- The canonical product-of-sums form for  $f_1$  is

$$\begin{aligned} f_1(a,b,c) &= M_0 \cdot M_3 \cdot M_5 \cdot M_7 \\ &= (a+b+c) \cdot (a+b'+c') \cdot \\ &\quad (a'+b+c') \cdot (a'+b'+c'). \end{aligned}$$

Observe that:  $m_j = M_j'$

| a | b | c | $f_1$ |
|---|---|---|-------|
| 0 | 0 | 0 | 0     |
| 0 | 0 | 1 | 1     |
| 0 | 1 | 0 | 1     |
| 0 | 1 | 1 | 0     |
| 1 | 0 | 0 | 1     |
| 1 | 0 | 1 | 0     |
| 1 | 1 | 0 | 1     |
| 1 | 1 | 1 | 0     |

# Shorthand: $\sum$ and $\prod$

- $f_1(a,b,c) = \sum m(1,2,4,6)$ , where  $\sum$  indicates that this is a sum-of-products form, and  $m(1,2,4,6)$  indicates that the minterms to be included are  $m_1$ ,  $m_2$ ,  $m_4$ , and  $m_6$ .
- $f_1(a,b,c) = \prod M(0,3,5,7)$ , where  $\prod$  indicates that this is a product-of-sums form, and  $M(0,3,5,7)$  indicates that the maxterms to be included are  $M_0$ ,  $M_3$ ,  $M_5$ , and  $M_7$ .
- Since  $m_j = M_j'$  for any  $j$ ,  
$$F_1(a,b,c) = \sum m(1,2,4,6) = \prod M(0,3,5,7)$$

# Conversion Between Canonical Forms

- Replace  $\sum$  with  $\prod$  (or *vice versa*) and replace those  $j$ 's that appeared in the original form with those that do not.

- Example:

$$\begin{aligned}f_1(a,b,c) &= a'b'c + a'bc' + ab'c' + abc' \\&= m_1 + m_2 + m_4 + m_6 \\&= \sum(1,2,4,6) \\&= \prod(0,3,5,7) \\&= (a+b+c) \cdot (a+b'+c') \cdot (a'+b+c') \cdot (a'+b'+c')\end{aligned}$$

# Standard Forms (NOT Unique)

- Standard forms are “*like*” canonical forms, except that not all variables need appear in the individual product (SOP) or sum (POS) terms.
- Example:  
$$f_1(a,b,c) = a'b'c + bc' + ac'$$

is a *standard* sum-of-products form
- $$f_1(a,b,c) = (a+b+c) \cdot (b'+c') \cdot (a'+c')$$

is a *standard* product-of-sums form.

# Conversion of SOP from standard to canonical form

- Expand *non-canonical* terms by inserting equivalent of 1 in each missing variable x:

$$(x + x') = 1$$

- Remove duplicate minterms

- $$\begin{aligned} f_1(a,b,c) &= a'b'c + bc' + ac' \\ &= a'b'c + (a+a')bc' + a(b+b')c' \\ &= a'b'c + abc' + a'bc' + abc' + ab'c' \\ &= a'b'c + abc' + a'bc + ab'c' \end{aligned}$$

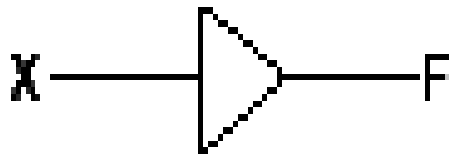
# Conversion of POS from standard to canonical form

- Expand noncanonical terms by adding 0 in terms of missing variables (*e.g.*,  $xx' = 0$ ) and using the distributive law
- Remove duplicate maxterms

- $$\begin{aligned} f_1(a,b,c) &= (a+b+c) \cdot (b'+c') \cdot (a'+c') \\ &= (a+b+c) \cdot (\textcolor{red}{a}a' + b' + c') \cdot (a' + \textcolor{red}{b}b' + c') \\ &= (a+b+c) \cdot (a+b'+c') \cdot (\textcolor{blue}{a}' + \textcolor{blue}{b}' + \textcolor{blue}{c}') \cdot \\ &\quad (a'+b+c') \cdot (\textcolor{blue}{a}' + \textcolor{blue}{b}' + \textcolor{blue}{c}') \\ &= (a+b+c) \cdot (a+b'+c') \cdot (a'+b'+c') \cdot (a'+b+c') \end{aligned}$$



# BUFFER



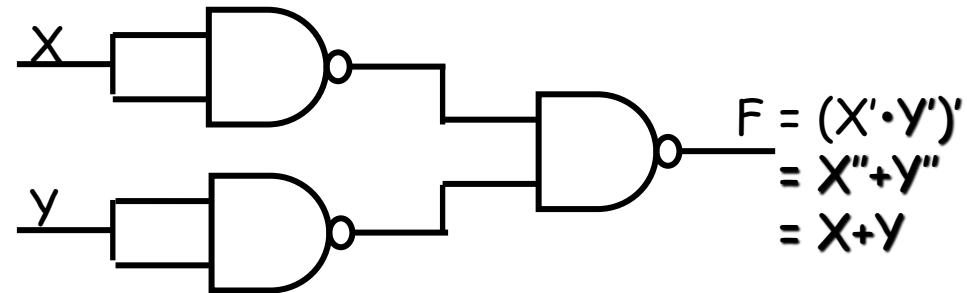
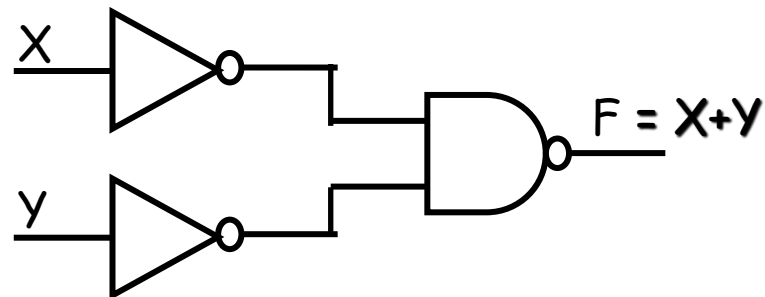
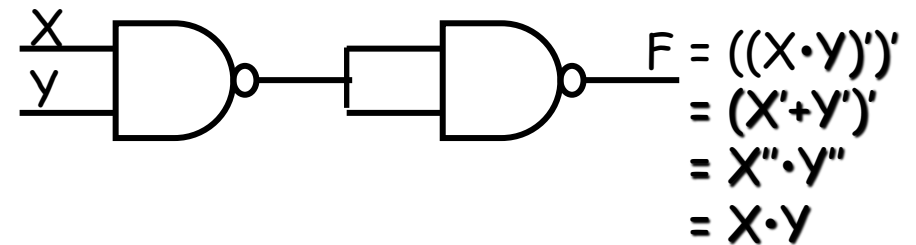
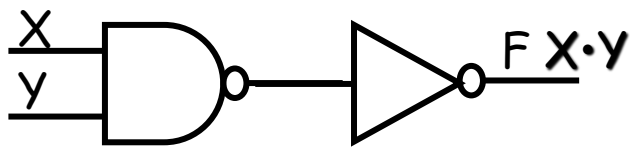
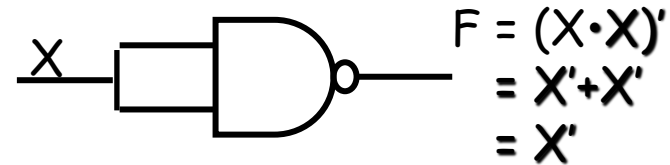
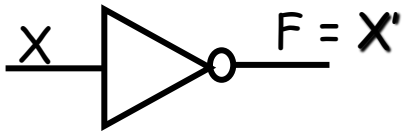
$$F = X$$

| X | F |
|---|---|
| 0 | 0 |
| 1 | 1 |

# NAND Gate

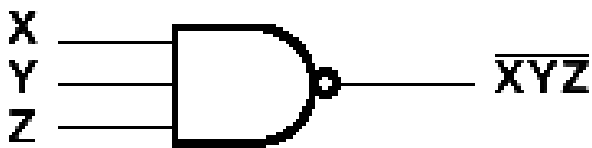
- Known as a “universal” gate because ANY digital circuit can be implemented with NAND gates alone.
- To prove the above, it suffices to show that AND, OR, and NOT can be implemented using NAND gates only.

# NAND Gate Emulation

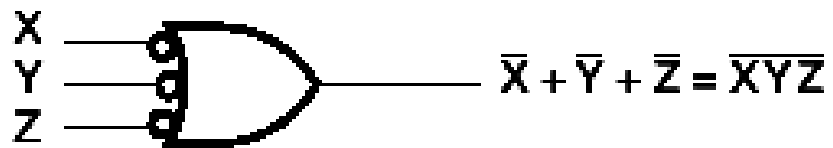


# NAND Circuits

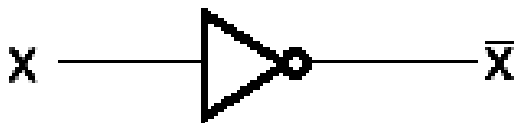
- To easily derive a NAND implementation of a Boolean function:
  - Find a simplified SOP
  - Design SOP with basic gates
  - Change AND-OR circuit to a NAND circuit using bubble
  - Use the alternative symbols below



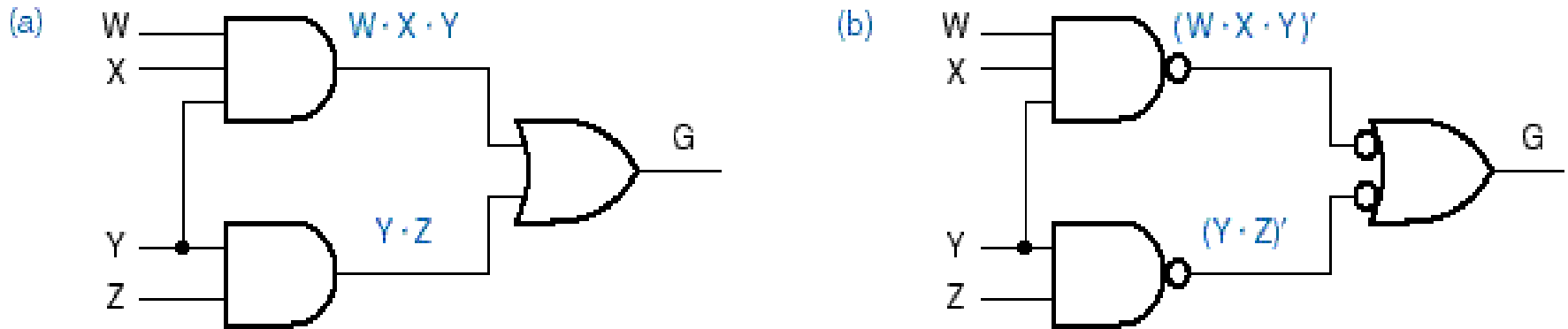
(a) AND - NOT



(b) NOT - OR



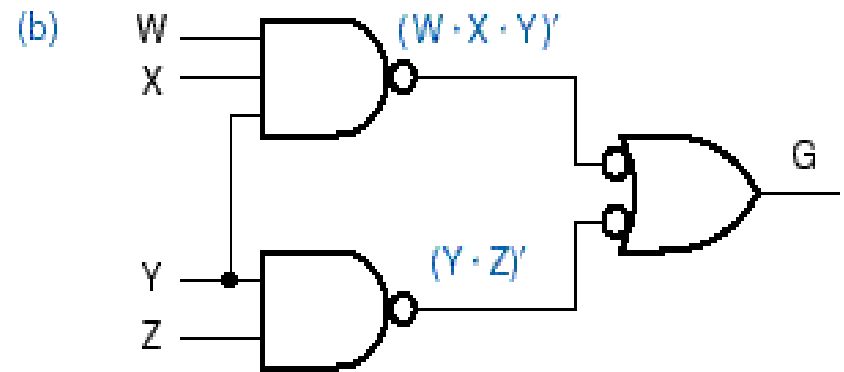
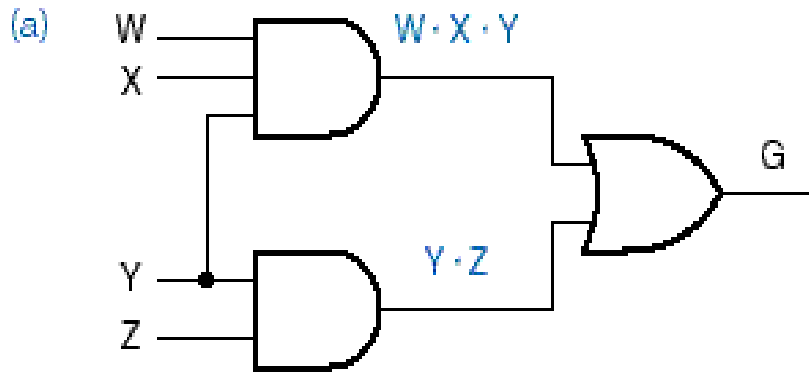
# AND-OR (SOP) Emulation Using NANDs



Two-level implementations

- a) Original SOP
- b) Implementation with NANDs

# AND-OR (SOP) Emulation Using NANDs (cont.)



**Verify:**

(a)  $G = WXY + YZ$

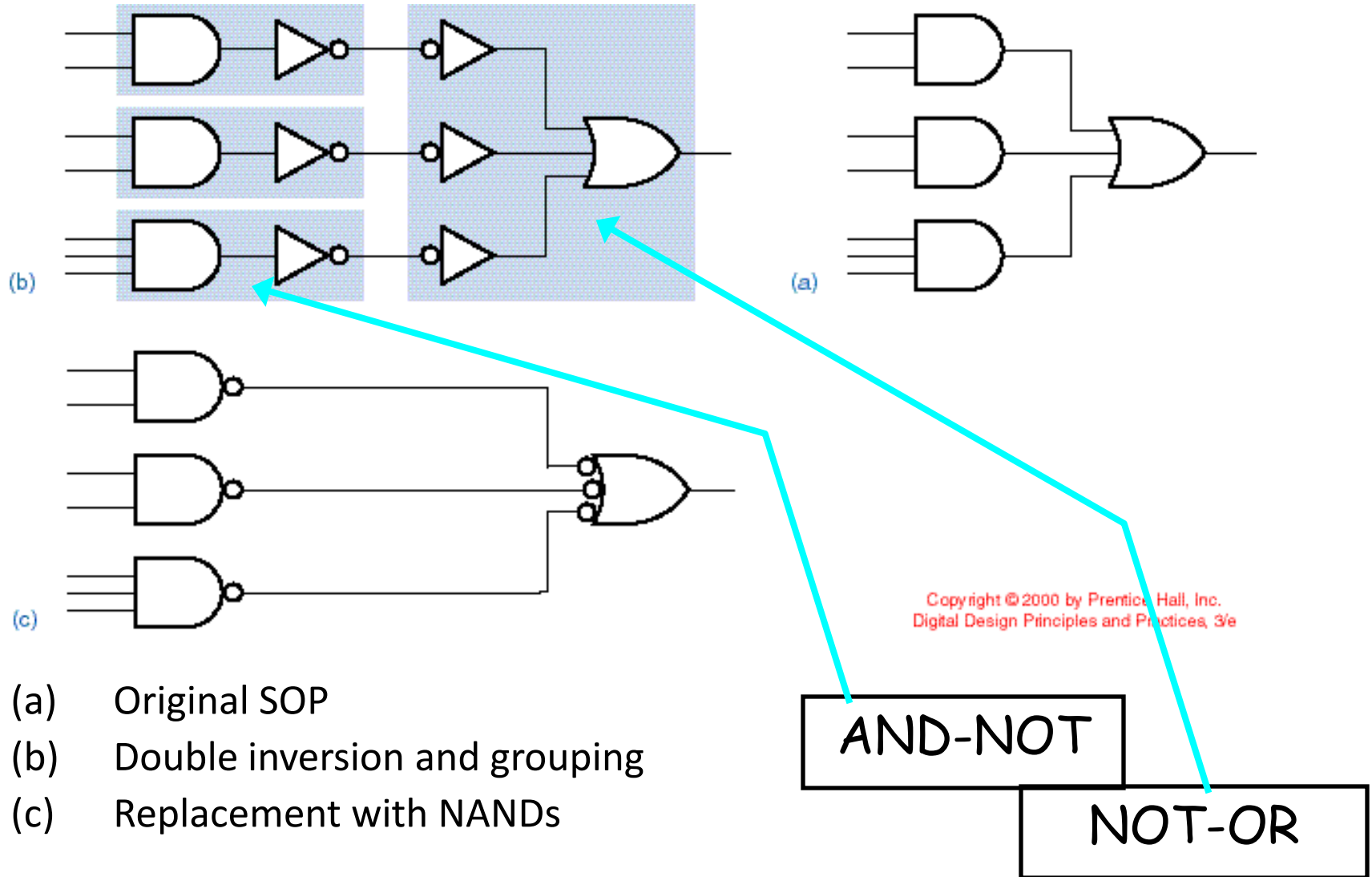
(b)  $G = ((WXY)' \cdot (YZ)')'$   
 $= (WXY)'' + (YZ)'' = WXY + YZ$

# Multilevel NAND Circuits

Starting from a multilevel circuit:

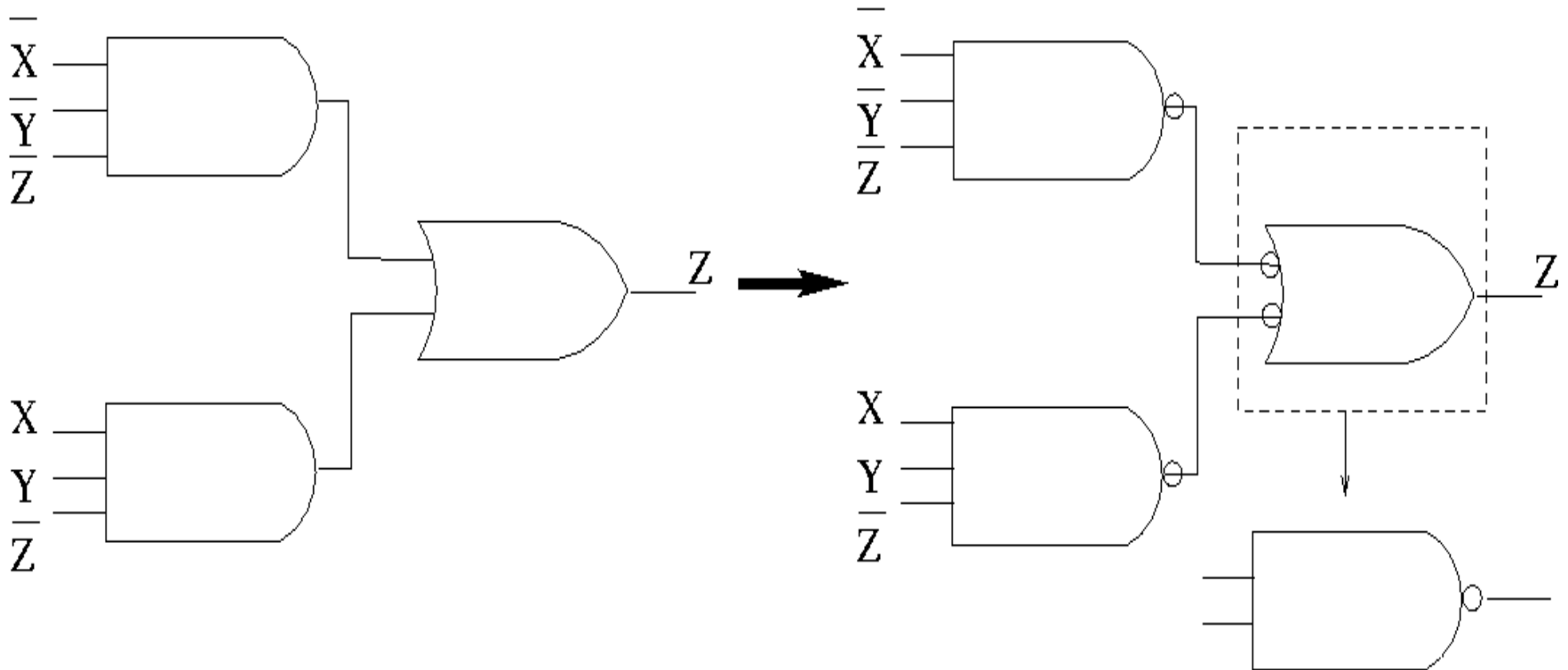
1. Convert all AND gates to NAND gates with AND-NOT graphic symbols.
2. Convert all OR gates to NAND gates with NOT-OR graphic symbols.
3. Check all the bubbles in the diagram. For every bubble that is not counteracted by another bubble along the same line, insert a NOT gate or complement the input literal from its original appearance.

# SOP with NAND





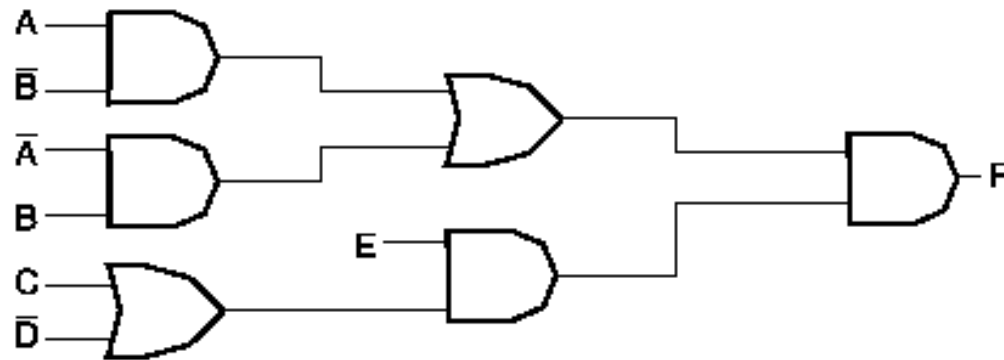
## Example (cont.)



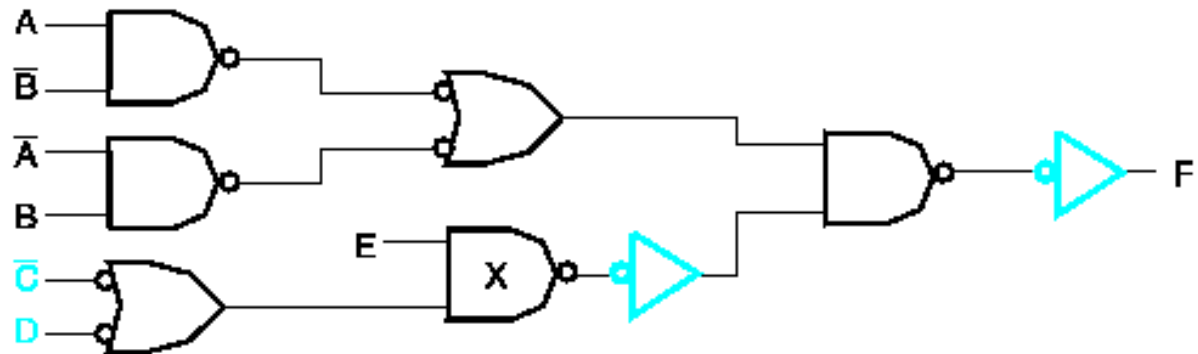
Two-level implementation with NANDs

$$F = X'Y'Z' + XYZ'$$

# Another Example!



(a) AND – OR gates



(b) NAND gates

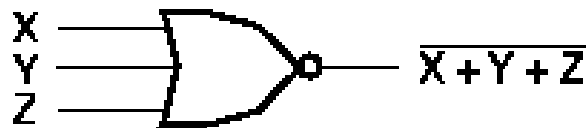
Fig. 2-32 Implementing  $F = (A\bar{B} + \bar{A}B)E(C + \bar{D})$

# NOR Gate

- Also a “universal” gate because ANY digital circuit can be implemented with NOR gates alone.
- This can be similarly proven as with the NAND gate.

# NOR Circuits

- To easily derive a NOR implementation of a Boolean function:
  - Find a simplified POS
  - Design POS using basic gates
  - Change OR-AND circuit to a NOR circuit
  - Use the alternative symbols below

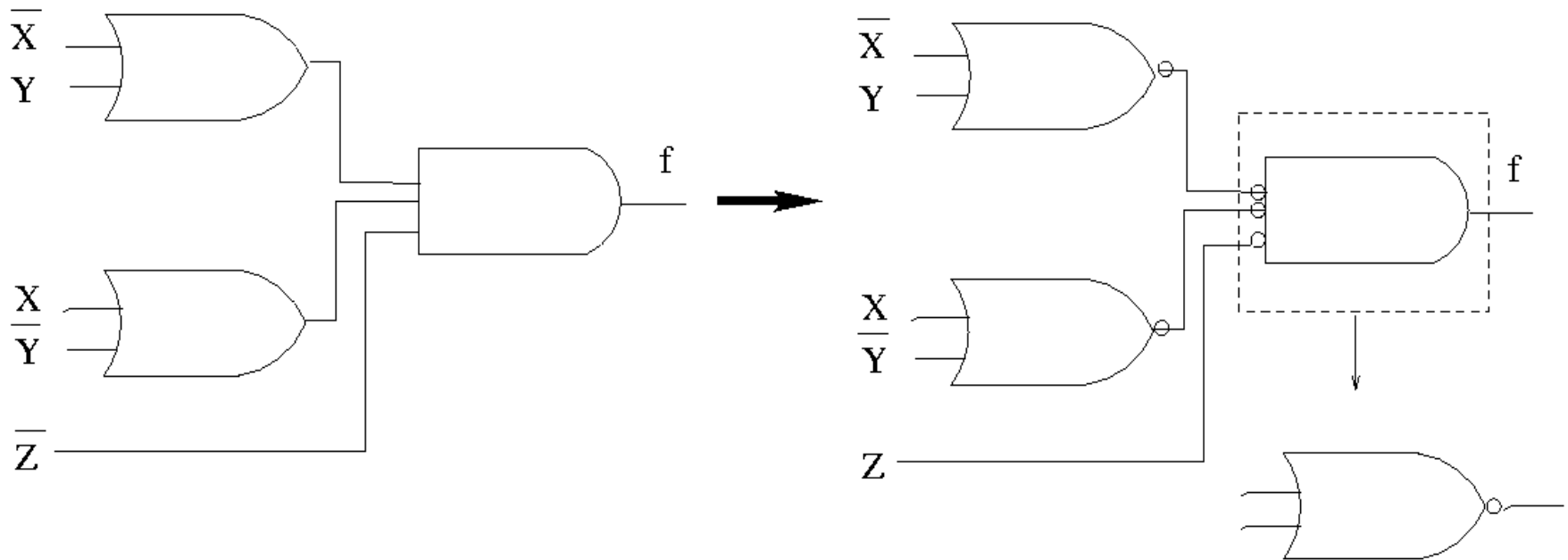


(a) OR – NOT



(b) NOT – AND

## Example (cont.)



Two-level implementation with NORs

$$F = (F')' = (X' + Y)(X + Y')Z'$$

# XOR function implementation

- $\text{XOR}(a,b) = ab' + a'b$
- Straightforward: 5 gates
  - 2 inverters, two 2-input ANDs, one 2-input OR
  - 2 inverters & 3 2-input NANDs
- Nonstraightforward:
  - 4 NAND gates

# XOR circuit with 4 NANDs

